

Students.xlsx X

EXP 5 > src > test > resources > Students.xlsx

	A	B	C	D
1	Roll No	Name	Marks	Grade
2	101	Student_1	76	B
3	102	Student_2	77	B
4	103	Student_3	78	B
5	104	Student_4	79	B
6	105	Student_5	80	A
7	106	Student_6	81	A
8	107	Student_7	82	A
9	108	Student_8	83	A
10	109	Student_9	84	A
11	110	Student_10	85	A

StudentUpdateTest.java 1 X

EXP 5 > src > test > java > com > experiment > StudentUpdateTest.java > ...

```
1  package com.experiment;
2
3  import org.apache.poi.ss.usermodel.Row;
4  import org.apache.poi.ss.usermodel.Sheet;
5  import org.apache.poi.ss.usermodel.Workbook;
6  import org.apache.poi.xssf.usermodel.XSSFWorkbook;
7  import org.junit.jupiter.api.BeforeAll;
8  import org.junit.jupiter.api.Test;
9
10 import java.io.FileInputStream;
11 import java.io.FileOutputStream;
12 import java.io.IOException;
13
14 import static org.junit.jupiter.api.Assertions.assertEquals;
15 import static org.junit.jupiter.api.Assertions.assertNotNull;
16 import static org.junit.jupiter.api.Assertions.assertTrue;
17
18 public class StudentUpdateTest {
19
20     private static final String FILE_PATH = "src/test/resources/Students.xlsx";
21
22     @BeforeAll
23     public static void createInitialExcelFile() throws IOException {
24         Workbook workbook = new XSSFWorkbook();
25         Sheet sheet = workbook.createSheet("Students");
26
27         Row header = sheet.createRow(0);
28         header.createCell(0).setCellValue("Roll No");
29         header.createCell(1).setCellValue("Name");
30         header.createCell(2).setCellValue("Marks");
31         header.createCell(3).setCellValue("Grade");
32
33         for (int i = 1; i <= 10; i++) {
34             Row row = sheet.createRow(i);
35             row.createCell(0).setCellValue(100 + i);
36             row.createCell(1).setCellValue("Student_" + i);
37             row.createCell(2).setCellValue(50.0);
38             row.createCell(3).setCellValue("C");
39         }
40     }
41 }
```

```

41  try (FileOutputStream fos = new FileOutputStream(FILE_PATH)) {
42      workbook.write(fos);
43  }
44  workbook.close();
45  }
46
47  @Test
48  public void testUpdateTenStudentRecords() throws IOException {
49      boolean updateSuccess = StudentRecordUpdater.updateStudentRecords(FILE_PATH);
50      assertTrue(updateSuccess, message: "File update operation failed");
51
52      try (FileInputStream fis = new FileInputStream(FILE_PATH);
53          Workbook workbook = new XSSFWorkbook(fis)) {
54
55          Sheet sheet = workbook.getSheetAt(0);
56
57          for (int i = 1; i <= 10; i++) {
58              Row row = sheet.getRow(i);
59              assertNotNull(row, "Row " + i + " should exist");
60
61              double actualMarks = row.getCell(2).getNumericCellValue();
62              String actualGrade = row.getCell(3).getStringCellValue();
63
64              double expectedMarks = 75.0 + i;
65              String expectedGrade = expectedMarks >= 80.0 ? "A" : "B";
66
67              assertEquals(expectedMarks, actualMarks, "Marks mismatch for record " + i);
68              assertEquals(expectedGrade, actualGrade, "Grade mismatch for record " + i);
69          }
70      }
71  }
72  }

```

```
● PS C:\Users\Afroz\Desktop\New folder\EXP 5> mvn test
[INFO] Scanning for projects...
[INFO] -----< com.experiment:student-record-updater >-----
[INFO] Building student-record-updater 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- resources:3.4.0:resources (default-resources) @ student-record-updater ---
[INFO] skip non existing resourceDirectory C:\Users\Afroz\Desktop\New folder\EXP 5\src\main\resources
[INFO] --- compiler:3.15.0:compile (default-compile) @ student-record-updater ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] --- resources:3.4.0:testResources (default-testResources) @ student-record-updater ---
[INFO] Copying 1 resource from src\test\resources to target\test-classes
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ student-record-updater ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] --- surefire:3.2.5:test (default-test) @ student-record-updater ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.experiment.StudentUpdateTest
ERROR StatusLogger Log4j2 could not find a logging implementation. Please add log4j-core to the classp
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.787 s -- in com.experiment.St
[INFO] Results:
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.776 s
[INFO] Finished at: 2026-08-08T16:41:08+05:30
[INFO] -----
```